



MILITARY INSTITUTE OF SCIENCE AND TECHNOLOGY

Project Proposal

KrishiChain

CSE-302: Database Management Systems Sessional

**Group - A6
Section-A**

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1. Introduction

Agriculture plays a vital role in Bangladesh's economy, yet farmers often face challenges such as unfair pricing, dependence on middlemen, inefficient storage management, and lack of transparent market access. These issues reduce farmers' profits and create an imbalance between producers and buyers.

KrishiChain is a Database Management System (DBMS)-based Agricultural Market Management System designed to provide a transparent, efficient, and technology-driven marketplace for agricultural products. The system introduces a region-based Virtual ARAT, where harvested crops are automatically assigned to the nearest regional market. Buyers can participate in competitive bidding, ensuring farmers receive fair market prices for their produce.

The system manages the complete lifecycle of agricultural products - from crop production and harvesting to storage management, bidding, transportation, payment processing, and daily market price monitoring. By maintaining centralized and reliable records, KrishiChain improves transparency, minimizes market manipulation, reduces transportation costs, and supports efficient decision-making for all stakeholders involved in the agricultural supply chain.

2. Project Objectives

The objectives of the project are:

- Develop a transparent digital agricultural marketplace.
- Enable farmers to sell crops directly through Virtual ARATs.
- Reduce dependency on middlemen and promote fair pricing.
- Automatically assign harvested crops to the nearest regional ARAT.
- Maintain accurate records of crop production, harvests, storage, transportation, and payments.
- Implement a secure bidding system for agricultural products.
- Monitor daily crop prices from different regional markets.
- Improve storage utilization through efficient harvest management.
- Maintain complete transaction history for future reporting and analysis.
- Provide a user-friendly software solution for farmers, buyers, transport personnel, and administrators.

3. Users

Farmer

- Register and manage harvested crops.
- Store harvested crops in available storage facilities.
- View bids placed by buyers.
- Accept successful bids.
- Receive payments after successful transactions.

Buyer

- Browse available harvest batches.
- Participate in bidding.
- Purchase crops through the bidding system.
- View transaction history.

Storage Manager

- Manage storage facilities.
- Monitor storage capacity.
- Allocate storage for harvest batches.
- Update storage status.

Transport Personnel

- View assigned transportation requests.
- Deliver purchased harvest batches.
- Update transportation status.
- Manage delivery records.

Administrator

- Manage users and user roles.
- Manage crop information.
- Manage Virtual ARATs.
- Monitor bidding activities.
- Manage daily market prices.
- Oversee storage, transportation, and payment records.
- Generate reports and maintain the overall system.

4. Front-End and Back-End Tools

Front-End

- **React.js** - Build a modern, responsive, and interactive user interface.
- **HTML5** - Structure the web pages.
- **CSS3** - Style the application.
- **Bootstrap 5 (or Tailwind CSS)** - Provide responsive design and reusable user-interface components.
- **JavaScript (ES6+)** - Implement client-side logic and interactivity.
- **Axios** - Support API communication between the front end and the back end.

Back-End

- **Node.js** - Provide the JavaScript runtime for server-side development.
- **Express.js** - Build RESTful APIs and manage server-side routes and requests.
- **Oracle Database 11g** - Serve as the primary relational database management system.
- **node-oracledb** - Connect the Node.js and Express.js back end with Oracle Database 11g.
- **SQL and PL/SQL** - Implement database queries, stored procedures, functions, triggers, and other database operations.

Development Tools

- Visual Studio Code
- Oracle SQL Developer
- Oracle Database 11g
- Postman - Test and validate APIs.
- Git
- GitHub

5. Features of the Project

User Management

- User Registration
- Secure Login
- Role-Based Access Control
- User Profile Management

Crop Management

- Add Crop Information
- Manage Crop Categories
- Update Base Prices
- Crop Information Maintenance

Harvest Management

- Record Harvest Batches
- Track Harvest Quantity
- View Harvest History
- Associate Harvests with Farmers

Virtual ARAT Management

- Region-Based Virtual ARAT
- Automatic Assignment to the Nearest ARAT
- View Available Harvest Batches
- Regional Marketplace Management

Storage Management

- Storage Registration
- Capacity Monitoring
- Harvest Storage Allocation
- Storage Availability Status

Bidding Management

- Buyers Place Bids
- Farmers View Multiple Bids
- Compare Offered Prices
- Accept Winning Bid
- Bid Status Tracking

Transportation Management

- Automatic Transport Assignment
- Driver Information Management
- Delivery Tracking

- Transportation Status Updates

Payment Management

- Payment Processing
- Payment History
- Transaction Record Maintenance
- Secure Payment Information

Daily Price Management

- Daily Crop Price Updates
- Regional Market Price Records
- Historical Price Tracking

Reporting Module

- Harvest Reports
- Storage Reports
- Sales Reports
- Payment Reports
- Daily Market Price Reports
- User Activity Reports